CYS103 Linear Algebra (2026/09)
TUTORIAL 1

Chapter 1: Linear Equations and Linear Transformations

1.1 System of Linear Equations

1.2 Echelon and Reduced Echelon Forms

1. Solve the following system using elementary row operations on the augmented matrix.

$$2x_1 + 4x_2 = -4$$
$$5x_1 + 7x_2 = 11$$

2. Solve the following systems.
   a.
$$x_2 + 4x_3 = -5$$
$$x_1 + 3x_2 + 5x_3 = -2$$
$$3x_1 + 7x_2 + 7x_3 = 6$$

CYS103 Linear Algebra (2026/09)
TUTORIAL 1

b. $$\begin{aligned} x_1 - 3x_2 &= 5 \\ -x_1 + x_2 + 5x_3 &= 2 \\ x_2 + x_3 &= 0 \end{aligned}$$

3. Determine if the following systems are consistent. (Do not completely solve the systems)

a. $$\begin{aligned} x_1 + 3x_3 &= 2 \\ x_2 - 3x_4 &= 3 \\ -2x_2 + 3x_3 + 2x_4 &= 1 \\ 3x_1 + 7x_4 &= -5 \end{aligned}$$

CYS103 Linear Algebra (2026/09)
TUTORIAL 1

b.
$$\begin{aligned} x_1 \qquad\qquad\quad - 2x_4 &= -3 \\ 2x_2 + 2x_3 \qquad\quad &= 0 \\ x_3 + 3x_4 &= 1 \\ -2x_1 + 3x_2 + 2x_3 + \quad x_4 &= 5 \end{aligned}$$

4. Determine which matrices are in reduced echelon form and which are in echelon form.

a. $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

b. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

c. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

d. $\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

CYS103 Linear Algebra (2026/09)
TUTORIAL 1

5. Row reduced the following augmented matrices into reduced echelon forms.
   a. $\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix}$

   b. $\begin{bmatrix} 3 & -4 & 2 & 0 \\ -9 & 12 & -6 & 0 \\ -6 & 8 & -4 & 0 \end{bmatrix}$